Systems Theory
Homework 4

1. Sketch the root loci for the pole-zeros configurations given in Figure 1.

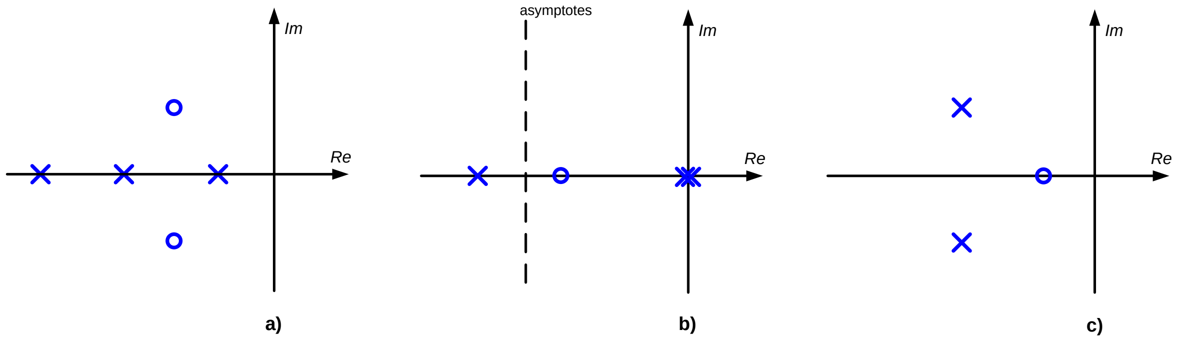

Figure 1. Pole zero configurations

2. Draw the root locus (including asymptotes, intersection with imaginary axis, break away point) of a system with the following open loop transfer function:

$$G(s) = \frac{K}{s(s+1)(s^2+4s+13)}$$

3. Space shuttle pitch control system

The problem considers only one axis of motion of a space shuttle when in orbit, which is around the y-axis (pitch) as shown in Figure 2. This type of motion is used when the shuttle, with its solar panels that are used to power the electrical equipment and to recharge the batteries, optimizes the sun's captured energy.

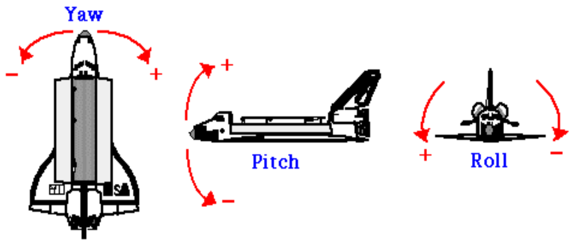

Figure 2 Space shuttle

The block diagram associated with the control system responsible for the pitch motion is given in Figure 3.

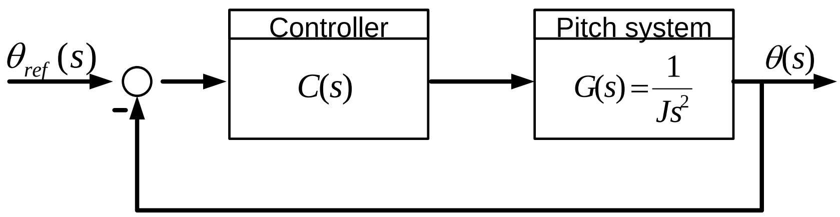

Figure 3 Pitch Control Block Diagram

The transfer function of the shuttle (subsystem responsible for the pitch movement) is:

$$G(s) = \frac{1}{Js^2}, \quad J = 12\, kgm^2,$$

$C(s)$ is the controller transfer function, $\theta(s)$ is the measured pitch angle (output) and $\theta_{ref}(s)$ is the desired (reference) pitch angle (input).

Analyze in the behavior of the space shuttle pitch control system (single axis control) from Figure 3 (check your results in Matlab).

1. Determine the closed-loop transfer function with various controllers ($K > 0$):
   - I. $C(s)$ is a proportional gain: $C(s) = K$
   - II. $C(s)$ is a Lead-Lag Compensator: $C(s) = \frac{K(s+1)}{s+2}$
2. Determine whether the system is stable in both cases (I, II). Analyze the influence of the gain K on system stability.
3. Draw the root locus of the control system from Figure 3 for both controllers and analyze the system stability and the location of the closed-loop poles.